

20251216 Back to nGIST

Make sure `mkdir \ngistTutorial\results` under `ngist_supplementary_public` first, so that at least the test command `ngistPipeline --config configFiles/MasterConfig.yaml --default-dir configFiles/defaultDir` can run successfully!!! We should see:

```
ngIST pipeline
[ DONE      ] ngIST completed successfully.
```

Before we start, checkout the `.yaml` and `defaultDir` files for current MAUVE internal release v2 at <https://www.canfar.net/storage/arc/list/projects/mauve/> and <https://ws-uv.canfar.net/arc/files/projects/mauve/products/configFiles/IC3392/IC3392.defaultDir>

Now my idea is to apply this setting to current `dev-branch` version of `ngIST` to see if it works. To begin, we still use the default testing dataset `NGC0000.fits`

After comparing with the provided `MasterConfig.yaml` and testing, below is the only parameter need to update in current new version of `ngIST`.

1. `LSF_DATA : 'lsf_MUSE-WFM'` => `LSF_DATA : 'lsf_MUSE-WFM-MILES'` #
*Path of the file specifying the line-spread-function of the observational data.
The specified path is relative to the configDir path in defaultDir.*
2. `LIBRARY : 'MILES_EMLINES/'` => `LIBRARY : 'MILES/'` # options are
'MILES/', 'miles_ssp_ch/', 'IndoUS', and 'Walcher/'
3. `LIBRARY : 'MILES_safe/'` => `LIBRARY : 'MILES/'` # options are
'MILES/', 'miles_ssp_ch/', or 'IndoUS/'

For the `LSF_DATA`, I compare two files and i confirm that that they just simply change the name. I guess that is because they want to distinguish it with the other LSF: `lsf_MUSE-WFM-UDF10`.

As for the `LIBRARY : 'MILES_EMLINES/'` and `LIBRARY : 'MILES_safe/'`, now i understand that they are two templates that we should put in the `spectralTemplates` folder and so I have downloaded them from OzStar. Clearly, they are the subset (in terms of both $[M/H]$ and $\log(\text{Age})$), but still unclear why those ranges (but I guess to first order speed up the fitting process) and the differences between them and how to determine the ranges? Will continue to check tomorrow.

Other parameters look fine for me. So far the testing dataset is working now.

So below is the complete config file for IC3392,
`MasterConfig_MAUVE_IC3392.yaml`:

```
# This is a test config file
GENERAL :
  RUN_ID : 'IC3392' # Name of the analysis run. A subdirectory
of this name within the output directory will be created. This
identifier further serves as a prefix to all output files.
  INPUT : 'IC3392_DATA_CUBE_FINAL_WCS_Pall_mad_red_v3.fits' #
Input file for this analysis run. The specified path is
relative to the input path given in defaultDir.
  OUTPUT : . # Output directory. The output of this run will
be collected in a subdirectory named RUN_ID. The specified
path is relative to the output path given in defaultDir.
  REDSHIFT : 0.005566 # Initial guess on the redshift of the
system [in z]. Spectra are shifted to rest-frame, according to
this redshift.
  PARALLEL: True # Use multiprocessing [True/False]
```

```

NCPU : 16 # Number of cores to use for multiprocessing
LSF_DATA : 'lsf_MUSE-WFM-MILES' # Path of the file
specifying the line-spread-function of the observational data.
The specified path is relative to the configDir path in
defaultDir.

OW_CONFIG : True # Ignore configurations from previous runs
which are saved in the CONFIG file in the output directory
[True/False]
OW_OUTPUT : True # Overwrite any output files already
present in the current output directory [True/False]
CREATE_HTML : False # option to create HTML pages similar to
Mapviewer outputs

# Read data module
READ_DATA :
    METHOD : 'MUSE_WFM' # Name of the routine in readData/
(without .py) to be used to read-in the input data.
    DEBUG : FALSE # Switch to activate debug mode [True/False]:
Pipeline runs on one, central line of pixels. Keep in mind to
clean output directory after running in DEBUG mode!
    ORIGIN : 218,218 # Origin of the coordinate system in pixel
coordinates: x,y (Indexing starts at 0).
    LMIN_TOT : 4800 # Spectra are shortened to the rest-frame
wavelength range defined by LMIN_TOT and LMAX_TOT. Note that
this wavelength range should be longer than all other
wavelength ranges supplied to the modules [in Angstroms.]
    LMAX_TOT : 7000
    LMIN_SNR : 4800 # Rest-frame wavelength range used for the
signal-to-noise calculation [in Angstroms.]
    LMAX_SNR : 7000
    EBmV : 0.0317 # Enter the E(B-V) value for the galaxy here
(or use null for no Galactic dust correction)

# Spatial masking module

```

SPATIAL_MASKING :

METHOD : 'default' # Name of the routine in spatialMasking/ (without .py) to perform the tasks. Set False to turn off module. Set 'default' to use the standard GIST implementation.

MIN_SNR : 1.5 # Spaxels below the isophote level which has this mean signal-to-noise level are masked.

MASK : 'IC3392_mask.fits' # File containing a spatial mask (Set False to not include a file).

Spatial binning module

SPATIAL_BINNING :

METHOD : 'voronoi' # Name of the routine in spatialBinning/ (without .py) to perform the tasks. Set False to turn off module. Set 'voronoi' to use the standard GIST implementation, exploiting the Voronoi tessellation routine of Cappellari & Copin (2003).

TARGET_SNR : 40.0 # Target signal-to-noise ratio for the Voronoi binning

COVARIANCE : 0.0 # Correct for spatial correlations of the noise during the Voronoi binning process according to the empirical equation $SNR \neq 1 + COVAR_VOR * np.log10(NSPAXEL)$ with NSPAXEL being the number of spaxels per bin (see e.g. Garcia-Benito et al. 2015).

Prepare spectra module

PREPARE_SPECTRA :

METHOD : 'default' # Name of the routine in prepareSpectra/ (without .py) to perform the tasks. Set False to turn off module. Set 'default' to use the standard GIST implementation.

VELSCALE : 53 # Spectral sampling of the logarithmically rebinned spectra [in km/s]; e.g. $velscale = d\lambda * C / \text{mean}(wave)$, with the spectral sampling of the linearly binned spectra $d\lambda$, the speed of light C , and the mean wavelength of the spectrum $\text{mean}(wave)$.

Stellar kinematics module

KIN :

METHOD : 'ppxf' # Name of the routine in stellarKinematics/ (without .py) to perform the tasks. Set False to turn off module. Set 'ppxf' to use the standard GIST implementation, exploiting the pPXF routine of Cappellari & Emsellem (2004).

#SPEC_PREMASK : 'specMask_KIN' # Optional mask if you want the first steps to use a different mask as compared to the final fit

SPEC_MASK : 'specMask_KIN' # File to define wavelength ranges to be masked during the stellar kinematics fit. The specified path is relative to the configDir path in defaultDir.

LMIN : 4800 # Rest-frame wavelength range used for the stellar kinematics analysis [in Angst.]

LMAX : 7000

SIGMA : 66 # Initial guess of the velocity dispersion of the system [in km/s]

BIAS : 'Auto' # Bias value for if MOM > 2. If set to 'Auto', bias is automaticalculated by Cappellari & Emsellem (2004):
bias = 0.7*np.sqrt(500./self.goodpixels.size)

MOM : 4 # Number of kinematic moments to be extracted

ADEG : 12 # Degree of the additive Legendre polynomial. Set '-1' to not include any additive polynomials

MDEG : 0 # Degree of the multiplicative Legendre polynomial. Set '0' to not include any multiplicative polynomials

MC_PPXF : 0 # Number of Monte-Carlo simulations to extract errors on the stellar kinematics. Formal errors are saved in any case.

LSF_TEMP : 'lsf_MILES' # Path of the file specifying the line-spread-function of the spectral templates. The specified path is relative to the configDir path in defaultDir.

TEMPLATE_SET : 'miles' # options are 'miles' or 'IndoUS' or 'walcher'

LIBRARY : 'MILES_EMLINES/' # options are 'MILES/', 'miles_ssp_ch/', 'IndoUS', and 'Walcher/'

```

    NORM_TEMP : 'LIGHT' # Normalise the spectral template
library to obtain light- or mass-weighted results [LIGHT /
MASS]

    DOCLEAN : True # use pPXF clean routine which performs 3-
sigma clipping based on the best-fit model [True / False]

    OPT_TEMP : 'default' # Keyword to select Optimal Template
method. Options are 'default', 'galaxy_single', 'galaxy_set'

    DUST_CORR : True # True/False, fixes the dust in the final
pPXF fit

    NOISE : 'variance' # Keyword to set noise to be used in pPXF
run. Option 'constant' and 'variance'

    PLOT : False # produce extra output plots for each bin

    DEBUG_BIN : [97] # Optional Keyword - array of bins [1,2,3]
- only works when parallel = False.

# Continuum Cube module
CONT :

    METHOD : 'ppxf' # Name of the routine in stellarKinematics/
(without .py) to perform the tasks. Set False to turn off
module. Set 'ppxf' to use the standard GIST implementation,
exploiting the pPXF routine of Cappellari & Emsellem (2004).

    #SPEC_PREMASK : 'specMask_KIN' # Optional mask if you want
the first steps to use a different mask as compared to the
final fit

    SPEC_MASK : 'specMask_KIN' # File to define wavelength
ranges to be masked during the stellar kinematics fit. The
specified path is relative to the configDir path in
defaultDir.

    LMIN : 4800 # Rest-frame wavelength range used for the
stellar kinematics analysis [in Angst.]

    LMAX : 7000

    SIGMA : 66 # Initial guess of the velocity dispersion of the
system [in km/s]

    MOM : 4 # Number of kinematic moments to be extracted

    ADEG : 8 # Degree of the additive Legendre polynomial. Set
'0' to not include any additive polynomials

```

```

# Do not include any additive polynomials
MDEG : 8 # Degree of the multiplicative Legendre polynomial.
Set '0' to not include any multiplicative polynomials
MC_PPXF : 0 # Number of Monte-Carlo simulations to extract
errors on the stellar kinematics. Formal errors are saved in
any case.
LSF_TEMP : 'lsf_MILES' # Path of the file specifying the
line-spread-function of the spectral templates. The specified
path is relative to the configDir path in defaultDir.
TEMPLATE_SET : 'miles' # options are 'miles' or 'IndoUS' or
'walcher'
LIBRARY : 'MILES_EMLINES/' # options are 'MILES/',
'miles_ssp_ch/', 'IndoUS', and 'Walcher/'
NORM_TEMP : 'LIGHT' # Normalise the spectral template
library to obtain light- or mass-weighted results [LIGHT /
MASS]
DOCLEAN : True # Keyword to turn on/off the sigma clipping.
Set to False for testing.
OPT_TEMP : 'default' # Keyword to select Optimal Template
method. Options are 'default', 'galaxy_single', 'galaxy_set'
DUST_CORR : True # True/False, fixes the dust in the final
pPXF fit
NOISE : 'variance' # Keyword to set noise to be used in pPXF
run. Option 'constant' and 'variance'
PLOT : False # produce extra output plots
DEBUG_BIN : [1] # Optional Keyword - array of bins [1,2,3] -
only works when parallel = False.
SAVE_HDF5 : False # Optional Keyword to save outputs in HDF5

# Emission line fitting module
GAS :
METHOD : 'ppxf' # options are 'ppxf', 'gandalf' or
'MAGPI_gandalf' Name of the routine in emissionLines/ (without
.py) to perform the tasks. Set False to turn off module. Set
'gandalf' to use the standard GIST implementation, exploiting
the pyGandALF routine of Sarzi et al. (2006).

```

```
SPEC_MASK : 'specMask_GAS'

LEVEL : 'BIN' # Set 'BIN' to perform the analysis bin-by-
bin, 'SPAXEL' for a spaxel-by-spaxel analysis, and 'BOTH' to
perform a spaxel-by-spaxel analysis that is informed by a
previous bin-by-bin analysis.

LMIN : 4800 # Rest-frame wavelength range used for the
emission-line analysis [in Angst.]

LMAX : 7000

ERRORS : 0 # Derive errors on the emission-line analysis (0
No / 1 Yes). Note: Due to limitations in pyGandALF, this is
currently not possible. We expect a new pyGandALF version to
be published soon.

REDDENING : False # Include the effect of reddening by dust
in the pyGandALF fit. Put in the form 0.1,0.1 without any
spaces. If you intend to use multiplicative polynomials
instead, set REDDENING to False and add a MDEG keyword in the
GAS section to set the polynomial order. Additive polynomials
cannot be used with pyGandALF.

MDEG : 8 # Degree of the multiplicative Legendre polynomial.
Set '0' to not include any multiplicative polynomials.

FIXED : True # Fix the stellar kinematics to the results
obtained with the stellar kinematics module [True / False]

MOM : 2 # Gas moments. Set to 2 for V and sigma. Higher
orders not yet tested

EBmV : null # As opposed to None. De-redden the spectra for
the Galactic extinction in the direction of the target
previously to the analysis. Use e.g. EBmV = A_v / 3.2

EMI_FILE : 'emissionLinesPHANGS.config' # Emission line set-
up file for emline fitter. The specified path is relative to
the configDir path in defaultDir. Set to
'emissionLinesPHANGS.config' when using the 'ppxf' routine,
and 'emissionLines.config' when using the 'gandalf' or
'MAGPI_gandalf' routines.

LSF_TEMP : 'lsf_MILES' # Path of the file specifying the
line-spread-function of the spectral templates. The specified
```



```

path is relative to the configDir path in defaultDir.

    TEMPLATE_SET : 'miles' # options are 'miles' or 'IndoUS'
    LIBRARY : 'MILES_EMLINES/' # options are 'MILES_EMLINES/'
    (contains a smaller subset of templates sim to that used by
    PHANGS) 'MILES/', 'miles_ssp_ch', or 'IndoUS'

    NORM_TEMP : 'LIGHT' # Normalise the spectral template
    library to obtain light- or mass-weighted results [LIGHT /
    MASS]

# Star formation histories module
SFH :
    METHOD : 'ppxf' # Name of the routine in
    starFormationHistories/ (without .py) to perform the tasks.
    Set False to turn off module. Set 'ppxf' to use the standard
    GIST implementation, exploiting the pPXF routine of Cappellari
    & Emsellem (2004).

    LMIN : 4800 # Rest-frame wavelength range used for the star
    formation histories analysis [in Angstroms.]

    LMAX : 7000

    #SPEC_PREMASK : 'specMask_SFH' # Optional mask if you want
    the first steps to use a different mask as compared to the
    final fit

    SPEC_MASK : 'specMask_SFH' # File to define wavelength
    ranges to be masked during the star formation histories
    analysis. The specified path is relative to the configDir path
    in defaultDir.

    SPEC_EMITCLEAN : False # Keyword (True/False) to
    enable/disable the use of emission line cleaned spectra from
    the Emission Line Fitting Module.

    MOM : 4 # Number of kinematic moments to be extracted. If
    you use FIXED = True, this should be same number of moments
    used to extract the stellar kinematics before. Otherwise the
    parameter can be set independently.

    MDEG : 12 # Degree of the multiplicative Legendre
    polynomial. Set '0' to not include any multiplicative
    polynomials. Note that additive Legendre polynomials cannot be

```

used for this module.

`REGUL_ERR` : 10000. # Regularisation error for the regularised run of pPXF. Note: Regularisation = $1 / \text{REGUL_ERR}$

`FIXED` : True # Fix the stellar kinematics to the results obtained with the stellar kinematics module [True / False]. If False, please provide an initial guess on the velocity dispersion of the systems [in km/s] by adding the parameter `SIGMA`.

`MC_PPXF` : 0 # Added to add the option of running MC on the SFH module

`LSF_TEMP` : 'lsf_MILES' # Path of the file specifying the line-spread-function of the spectral templates. The specified path is relative to the configDir path in defaultDir.

`TEMPLATE_SET` : 'miles' # options are 'miles' or 'IndoUS (not recc.)'

`LIBRARY` : 'MILES_safe/' # options are 'MILES/', 'miles_ssp_ch/', or 'IndoUS/'

`NORM_TEMP` : 'LIGHT' # Normalise the spectral template library to obtain light- or mass-weighted results [LIGHT / MASS]

`DOCLEAN` : True # use pPXF clean routine which performs 3-sigma clipping based on the best-fit model [True / False]

`OPT_TEMP` : 'default' # Keyword to select Optimal Template method. Options are 'default', 'galaxy_single', 'galaxy_set'

`DUST_CORR` : True # True/False, fixes the dust in the final pPXF fit

`NOISE` : 'variance' # Keyword to set noise to be used in pPXF run. Option 'constant' and 'variance'

`PLOT` : False # produce extra output plots

`DEBUG_BIN` : [1] # Optional Keyword - array of bins [1,2,3] - only works when parallel = False.

Line strenghts module

`LS` :

`METHOD` : 'default' # Name of the routine in lineStrenghts/ (without .py) to perform the tasks. Set False to turn off

(without .py) to perform the tasks. See `help` to learn on module. Set 'default' to use the standard GIST implementation, exploiting the routines of Kuntschner et al. (2006) and Martin-Navarro et al. (2018).

`TYPE` : 'SPP' # Set 'LS' to only measure line strength indices, or 'SPP' to also match these indices to stellar population properties.

`LS_FILE` : 'lsBands.config' # File to define the wavelength band of the line strength indices to be measured. The specified path is relative to the configDir path in defaultDir.

`CONV_COR` : 8.4 # Spectral resolution [in Angst.] at which the line strength indices are measured.

`SPP_FILE` : 'MILES_KB_LIS8.4.fits' # File providing predictions on line strength indices for a set of single stellar population models

`MC_LS` : 30 # Number of Monte-Carlo simulations in order to obtain errors on the line strength indices. Note: This must be turned on.

`NWALKER` : 200 # Number of walkers for the MCMC algorithm (used for the conversion of indices to population properties)

`NCHAIN` : 200 # Number of iterations in the MCMC algorithm (used for the conversion of indices to population properties)

`LSF_TEMP` : 'lsf_MILES' # Path of the file specifying the line-spread-function of the spectral templates. The specified path is relative to the configDir path in defaultDir.

`TEMPLATE_SET` : 'miles' # options are 'miles' or 'IndoUS'

`LIBRARY` : 'MILES_safe/' # options are 'MILES/', 'miles_ssp_ch/', or 'IndoUS/'

`NORM_TEMP` : 'LIGHT' # Normalise the spectral template library to obtain light- or mass-weighted results [LIGHT / MASS]

User modules

`UMOD` :

`METHOD` : False # Name of the routine in User Module /

(without .py) to perform the tasks. Set False to turn off module. More info soon